

MATHGUARD

A policy gateway for AI agents

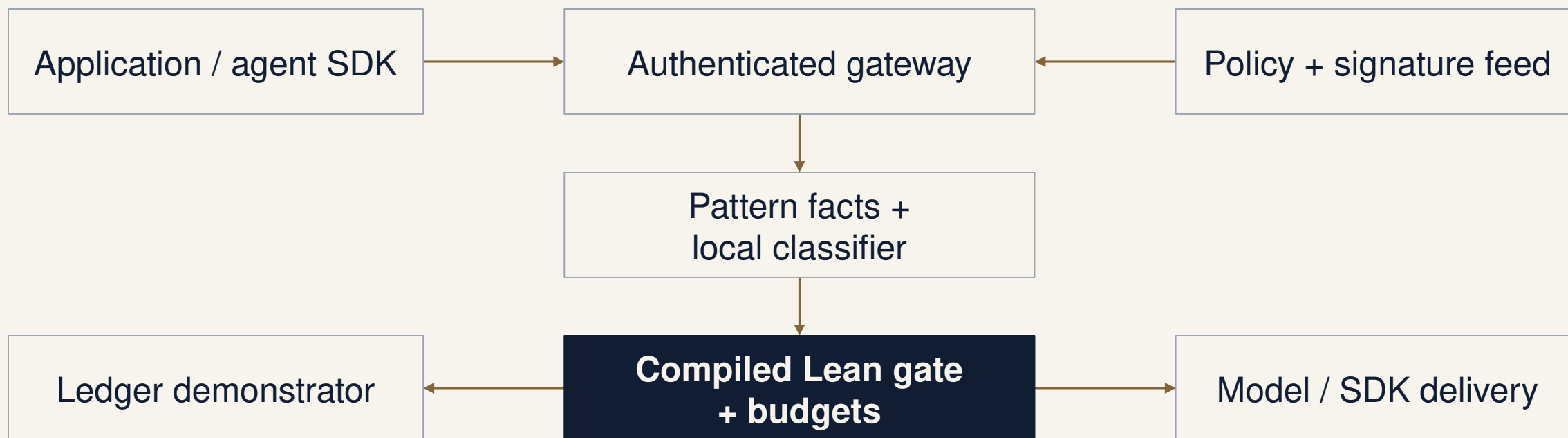
Inspect prompts, tool calls and messages.
Apply policy before releasing content
or committing a tool effect.

TEAM MATHGUARD



One enforcement path

Applications route their AI interactions through the same gate.



Sanitized decisions and resources feed the dashboard and audit export.

Hard restrictions remain in force

A semantic “safe” result cannot override a deterministic denial.

DETERMINISTIC FACTS	SEMANTIC RESULT	FINAL ADMISSION
Signature matched	Safe / allow	Denied
Tool outside allowlist	Safe / allow	Denied
No hard restriction	At or above block threshold	Denied + review
Classifier unavailable	Balanced policy	Review, no dispatch

Deny, review and redact flags combine by union.

Proofs cover rule composition. Detector completeness remains empirical.

Live policy and signature edits

Observed through the public HTTP rehearsal with fixture verdicts.

HTTP REHEARSAL	ACTIVE VERSION	RESULT
Add a synthetic signature	Feed 2	Matching content blocked
Remove that signature	Feed 3	Same content allowed
Tighten PII handling	Policy 8	Email now blocked
Submit invalid policy JSON	Policy 8 retained	Last valid policy kept

Activation preserves balances and previously charged resources.

Resource governance before dispatch

01 **Reserve a configured upper bound**

Money, tokens, compute time and call count are charged conservatively.

02 **Stop exhausted budgets and runaway sessions**

Actual worker tests reject before dispatch.
Per-session step and repeat caps apply.

03 **Quarantine after provider timeout**

The adapter stops. An operator confirms upstream cleanup before recovery.

Ledger approval and replay

Synthetic demonstrator accounts. Balances in PLN.

OBSERVED TRANSITION	ALICE (PLN)	BOB (PLN)
Initial state	1,000.00	200.00
25.00 transfer	975.00	225.00
100.00 approved transfer	875.00	325.00
Exact retries	875.00	325.00

Approval issuance causes no payment. Exact retry adds no debit or charge.

Attack and artifact boundaries

01 **Indirect and encoded content**

Shared controls inspect tool results and session history.
Encoded sensitive data is refused.

02 **Pinned inert artifacts**

Bounded byte, hash and header checks.
Pickle and code-running formats are refused.

03 **Finite detector coverage**

Synthetic feed examples and labeled fixtures
demonstrate the enforcement paths.

Reproducible evidence

83

PASSED

Gateway / provider / SDK / evaluator suite

Actual worker + fixtures

Live-model accuracy and independent unseen evaluation remain pending.

156 is the audit catalog size. It does not count distinct security requirements.

EVIDENCE	RESULT	SCOPE
Fresh axiom audit	156 records	Pinned Lean + Mathlib
Static regressions	14 passed	Audit / parser
Public HTTP rehearsal	16 passed	Policy, feed, approval, export

Reporting for operators and judges

01 **Decision and resource visibility**

Dashboard: allow, block, review and redaction.
Policy/feed versions, p50/p95 and remaining bounds.

02 **Sanitized management export**

The HTTP rehearsal verified that example PII
and credentials are absent from exported metadata.

03 **Explicit assurance and current limits**

Fixture/live mode, quarantine and invalid reloads.
The volatile audit window remains visible.

Demonstration and submission

```
make test
```

```
make run
```

LOCAL MODEL RECORD

Exact installed model ID and endpoint
Server / weights version and license
Live evaluator results

github.com/ljaniec/Mathguard/pull/5

REMAINING EVIDENCE

Independent human-authored
prompt set

Browser / operator rehearsal
Registered member list
HackTribe entry